

A PROJECT BASED SEMINAR REPORT

on

Cognix

**Bayesian Multi-Agent Decision System with Calibrated Uncertainty
Quantification**

Submitted towards the
Partial Fulfilment of the Requirements of

**Bachelor of Technology in
Artificial Intelligence and Data Science**

By

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CERTIFICATE

This is to certify that the seminar report entitled

Cognix

**Bayesian Multi-Agent Decision System with Calibrated Uncertainty
Quantification**

being submitted by

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is a bonafide work carried out by student under the supervision of Ifat. J. Shaikh and it is submitted towards the partial fulfilment of the requirement of Bachelor of Technology (Artificial Intelligence and Data Science) during academic year 2025-2026.

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Abstract

Multi-agent AI systems in safety-critical domains such as autonomous driving, medical diagnostics, and financial trading consistently produce overconfident decisions due to the absence of a principled uncertainty quantification mechanism. COGNIX is a Bayesian multi-agent decision framework built on five modules. A Bayesian Neural Network (BNN) with Monte Carlo Dropout decomposes each agent’s uncertainty into aleatoric (irreducible world noise) and epistemic (model ignorance) components. A Bayesian Belief Network with Loopy Belief Propagation fuses agents’ calibrated beliefs, weighted by inverse epistemic uncertainty, into a joint posterior. Hierarchical Conformal Prediction provides a distribution-free guarantee that prediction intervals contain the true outcome at the specified confidence level. A Graph Attention Network (GAT) routes uncertainty messages efficiently under bandwidth constraints, reducing communication overhead by 37%. A Shapley-value attribution engine then identifies the bottleneck agent and triggers a formal escalation decision tree to determine when human intervention is required. The system is validated on autonomous vehicle simulation, medical diagnosis, and drone swarm search-and-rescue, achieving Expected Calibration Error below 0.05 and escalation F1 above 0.80 within a 100ms real-time budget.

Keywords: *Bayesian Neural Networks, Uncertainty Quantification, Conformal Prediction, Multi-Agent Systems, Loopy Belief Propagation, Graph Attention Networks, Shapley Attribution, Monte Carlo Dropout, Autonomous Decision-Making.*

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Chapter 1

Introduction

1.1 Project Idea

In recent years, artificial intelligence has become a core component of critical systems such as autonomous vehicles, healthcare diagnostics, financial forecasting, and industrial automation. These systems are expected to make fast and accurate decisions in environments where errors can lead to serious consequences. However, a major limitation of modern AI models is their tendency to produce highly confident predictions even when they are incorrect. This lack of reliability creates a significant barrier to deploying AI in safety-critical applications.

Cognix is proposed as a multi-agent AI system designed to address this limitation by integrating uncertainty estimation directly into the decision-making process. Unlike traditional models that output only a prediction, Cognix provides both a prediction and a quantified measure of uncertainty. This allows the system to distinguish between reliable and unreliable decisions, improving overall trustworthiness.

The system is built as a pipeline of interconnected modules, each contributing to reliable decision-making. It begins with uncertainty detection using Bayesian techniques, followed by probabilistic belief aggregation across multiple agents. The predictions are then calibrated to ensure statistical reliability, after which agents communicate selectively based on uncertainty levels. Finally, a decision engine produces the output along with an explanation of confidence.

The key technologies that power Cognix include:

- **Bayesian Neural Networks (BNN):** Used for decomposing uncertainty into aleatoric and epistemic components, enabling a deeper understanding of prediction reliability.

-
- **Multi-Agent Systems:** Multiple agents independently process data and collaborate to improve robustness and reduce single-point failure.
 - **Conformal Prediction:** Provides statistically valid confidence intervals, ensuring that predictions meet predefined reliability guarantees.
 - **Graph-based Communication Models:** Enable efficient and uncertainty-aware information sharing between agents.

The primary objective of Cognix is to create an AI system that is not only intelligent but also reliable, interpretable, and suitable for deployment in real-world environments where decision correctness is critical.

1.2 Motivation

The development of Cognix is driven by the increasing reliance on artificial intelligence in real-world applications and the risks associated with unreliable or overconfident predictions. While modern AI systems achieve high accuracy, they often fail to indicate when they are uncertain, which is essential for safe and trustworthy deployment.

The key factors motivating this project are:

1. **Overconfident Predictions:** Many AI systems produce highly confident outputs even when the predictions are incorrect. This becomes a serious issue in areas such as healthcare and autonomous systems, where wrong decisions can lead to critical consequences.
2. **Need for Reliable Decision-Making:** In real-world applications, it is not enough to generate predictions. Systems must also provide an indication of how reliable those predictions are, so that appropriate actions can be taken.
3. **Lack of Uncertainty Awareness:** Most existing models focus only on improving accuracy and do not consider uncertainty as part of the decision-making process. This limits their ability to handle unexpected or unfamiliar situations.
4. **Complexity of Multi-Agent Systems:** Modern applications involve multiple agents working together, such as in autonomous vehicles or distributed sensor networks. These systems require proper handling of uncertainty across all agents to make consistent and safe decisions.
5. **Advancements in Technology:** With the availability of modern machine learning frameworks, computational power, and simulation tools, it is now possible to design systems that can explicitly model and use uncertainty in decision-making.

The motivation behind Cognix is to develop a system that not only improves prediction accuracy but also makes the decision-making process more reliable, transparent, and suitable for real-world deployment.

1.3 Literature Survey

The development of Module 2 is informed by research on Bayesian uncertainty quantification, Bayesian Belief Networks in distributed systems, and Bayesian Networks for explainable probabilistic reasoning. The following papers are directly relevant to the Bayesian Belief Network module and are used as the research foundation for this report.

The first paper reviewed is “*A Bayesian Data-Driven Framework for Aleatoric and Epistemic Uncertainty Quantification in Remaining Useful Life Predictions*” by Jiang, Xing, Zio, and Zhu, published in *IEEE Sensors Journal* in 2024 [?]. The paper proposes a Bayesian framework that separates aleatoric uncertainty from epistemic uncertainty using Monte Carlo sampling in a relevance vector machine model. Its main contribution is the clear decomposition of total predictive uncertainty into uncertainty caused by inherent data noise and uncertainty caused by insufficient model knowledge. This is important for Module 2 because the belief network must not treat all uncertainty equally. The epistemic component is used as the reliability signal during message passing. The limitation of this work is that it focuses on a single predictive model in remaining useful life prediction and does not address multi-agent belief fusion, iterative message passing, or graph-level posterior formation. Module 2 extends the usefulness of this uncertainty decomposition by using epistemic uncertainty as a weighting factor inside a multi-agent Bayesian graph.

The second paper is “*A Bayesian Belief Network Framework for Protecting Generative AI-driven Attack in Smart Grid Communication*” by Munir, Proddatoori, Muralidhara, and Shetty, published in the 2025 Workshop on Computing, Networking and Communications [?]. This work develops a Bayesian Belief Network framework to analyze uncertain and adversarial behaviour in smart grid communication. The authors construct BBN nodes from operational parameters, use Pearson correlation to initialize dependencies, and compute mutual information to identify the most influential attack features. This paper is directly relevant because it demonstrates that BBNs are practical for distributed, safety-critical environments where several parameters interact under uncertainty. However, the work is domain-specific to smart grid cyberattack explanation and does not perform agent-to-agent belief propagation using epistemic uncertainty. Module 2 adopts the idea of dependency-aware graph construction but applies it to collaborative agents and uses Loopy Belief Propagation for collective posterior inference.

The third paper is “*Integrating Knowledge Graphs and Bayesian Networks: A Hybrid Approach for Explainable Disease Risk Prediction*” by Nzomo and Moodley, published in *IEEE COMPSAC 2025* [?]. The paper constructs Bayesian Networks from ontology-based knowledge graphs and multimodal electronic health records to support explainable disease risk prediction. It shows that Bayesian Networks can combine structured domain knowledge with data-driven probability estimation while still producing interpretable probabilistic risk assessments. This is important for Module 2 because Cognix also requires transparent

reasoning under incomplete and uncertain evidence. The limitation is that the proposed network is primarily a disease risk prediction system and does not focus on real-time multi-agent consensus or iterative belief stabilization. Module 2 adapts the explainability and structured dependency principles from this paper into a dynamic multi-agent setting.

Together, these papers establish the research basis for Module 2. The first paper justifies using epistemic uncertainty as a reliability signal, the second demonstrates the use of BBNs for uncertain distributed environments and dependency explanation, and the third supports the use of Bayesian Networks for explainable probabilistic reasoning. The identified research gap is the absence of a real-time multi-agent belief fusion module that combines uncertainty-aware weighting, correlation handling, iterative message passing, and convergence-based posterior computation. The Bayesian Belief Network module of Cognix is designed to fill this gap.

Chapter 2

Problem Definition and Scope

2.1 Problem Statement

Artificial Intelligence systems are increasingly used in critical domains such as autonomous vehicles, healthcare diagnostics, financial decision-making, and multi-robot systems. While these systems have achieved high accuracy, a major limitation still exists — they often produce decisions without properly representing uncertainty. As a result, systems may take incorrect actions with high confidence, which can lead to serious real-world consequences.

Another key challenge arises in multi-agent environments, where multiple AI systems or sensors contribute to a shared decision. In such settings, there is no unified mechanism to quantify, combine, and interpret uncertainty across different agents. Existing approaches typically focus on single-model predictions and fail to address how uncertainty should be communicated, calibrated, and used for decision-making in a distributed system.

Therefore, there is a need to design a system that not only generates predictions but also provides reliable uncertainty estimates, combines information from multiple agents, ensures calibrated confidence, and supports explainable decision-making. The Cognix system aims to address this gap by developing a structured framework for uncertainty-aware multi-agent decision systems.

2.1.1 Goals and Objectives

- **Goals**

1. Develop Cognix as a unified multi-agent decision system that incorporates uncertainty awareness at every stage of processing.
2. Enable reliable and interpretable decision-making by integrating prediction, uncertainty estimation, calibration, and explanation within a single framework.

3. Improve safety and robustness of AI systems operating in real-world environments by reducing overconfident incorrect decisions.
4. Build a scalable architecture that supports multiple agents and heterogeneous data sources such as sensors, simulations, and real-world inputs.

• Objectives

1. **Framework Development:** Design and implement a complete 5-module Bayesian multi-agent system for end-to-end uncertainty quantification and decision-making.
2. **Uncertainty Modeling:** Decompose uncertainty into aleatoric (irreducible data noise) and epistemic (reducible model uncertainty) at the individual agent level.
3. **Confidence Calibration:** Provide mathematically guaranteed and reliable confidence intervals using conformal prediction techniques.
4. **Efficient Communication:** Enable intelligent and uncertainty-aware communication between agents using Graph Neural Networks (GNNs).
5. **Uncertainty Attribution:** Identify and quantify the contribution of each agent to overall system uncertainty using Shapley value-based methods.
6. **Multi-Domain Validation:** Evaluate system performance across real-world domains such as autonomous driving (CARLA), medical diagnosis (CheXpert), and drone swarm search rescue.
7. **Regulatory Compliance:** Ensure the system meets modern AI safety and transparency standards such as ISO 26262, EU AI Act 2.0, and FDA guidelines.

2.1.2 Assumptions and Scope

• Assumptions

- ◇ **Data Availability:** It is assumed that sufficient training and testing data is available through simulation environments and real-world datasets.
- ◇ **Sensor Inputs:** The system assumes that agents provide structured input data such as images, numerical readings, or extracted features suitable for machine learning models.
- ◇ **Computational Resources:** It is assumed that adequate computational resources are available for model training and real-time inference.
- ◇ **Model Generalization:** The system assumes that models trained on simulation data can generalize reasonably well to real-world scenarios.

• Scope

-
- ◊ The project focuses on developing a multi-module AI system that integrates uncertainty estimation, probabilistic reasoning, calibration, communication, and decision-making.
 - ◊ The system is designed for multi-agent environments such as autonomous vehicles, robotics, and distributed sensing systems.
 - ◊ Validation is carried out using simulation platforms and controlled experimental setups.
 - ◊ The project emphasizes system design, model implementation, and evaluation rather than large-scale industrial deployment.
 - ◊ The system outputs include predictions along with uncertainty measures and explanations for decision-making.
 - ◊ User interface development and large-scale production deployment are not included in the scope of this project.

By clearly defining the problem, goals, and scope, the Cognix system aims to provide a reliable and structured approach for uncertainty-aware decision-making in modern AI systems.

Chapter 3

Methodology

3.1 Methodology

Module 2 – the Bayesian Belief Network – receives the structured output of Module 1 and performs collective belief fusion. The methodology is based on probabilistic graphical modelling and Loopy Belief Propagation. Each agent is represented as a node in a Bayesian graph, each node maintains a belief distribution, and messages are passed between connected agents until the system reaches a stable collective posterior.

3.1.1 Belief Representation

Each agent begins with its own local belief derived from its observation and uncertainty tuple:

$$O_i = (\mu_i, \sigma_{a,i}, \sigma_{e,i}) \quad (3.1)$$

where μ_i is the predicted mean or risk probability, $\sigma_{a,i}$ is aleatoric uncertainty, and $\sigma_{e,i}$ is epistemic uncertainty. For probability-valued beliefs, Module 2 represents each agent's local belief using a Beta distribution:

$$B_i \sim \text{Beta}(\alpha_i, \beta_i) \quad (3.2)$$

The Beta distribution is selected because it is naturally defined over $[0, 1]$, making it suitable for risk probabilities, confidence scores, and binary event likelihoods such as “safe” versus “unsafe” or “disease present” versus “disease absent.”

3.1.2 Graph Construction

The Bayesian Belief Network is represented as a graph $G = (V, E)$ where each node $v_i \in V$ denotes an agent and each edge $e_{ij} \in E$ denotes an information-sharing relationship or dependency between agents. The graph can be built from communication links, sensor overlap, domain dependency, or empirical correlation. This graph structure allows the module to reason not only over individual predictions but also over how those predictions influence each other.

3.1.3 Message Passing

Agents exchange messages with their neighbours. Each message contains the sender's current belief distribution, reliability score, epistemic uncertainty, aleatoric uncertainty, and correlation-adjusted influence weight. A message from agent j to agent i can be represented as:

$$m_{j \rightarrow i} = (B_j, r_j, \sigma_{a,j}, \sigma_{e,j}, \rho_{ij}) \quad (3.3)$$

where r_j is the reliability score and ρ_{ij} is the correlation factor between agents i and j .

3.1.4 Epistemic-Uncertainty-Weighted Influence

The key design decision in Module 2 is that agent influence is inversely related to epistemic uncertainty. If an agent is unsure because its model lacks knowledge, its message should contribute less to the collective posterior. The influence weight is computed as:

$$w_{j \rightarrow i} = \frac{r_j}{\sigma_{e,j} + \epsilon} \times (1 - \rho_{ij}) \quad (3.4)$$

where ϵ is a small constant that prevents division by zero. The term $(1 - \rho_{ij})$ reduces the weight of agents whose evidence is highly correlated with the receiver or with other incoming messages. This helps prevent double counting of redundant evidence.

3.1.5 Belief Update

At each iteration, every agent updates its belief by combining its current belief with incoming neighbour messages. The update follows the intuition that reliable and low-epistemic-uncertainty agents should exert stronger influence:

$$B_i^{(t+1)} = \eta \left(B_i^{(t)} \prod_{j \in N(i)} m_{j \rightarrow i}^{w_{j \rightarrow i}} \right) \quad (3.5)$$

where $N(i)$ is the set of neighbours of agent i and η is a normalization constant. After update, the module recomputes each node's posterior belief parameters and prepares the next message-passing iteration.

3.1.6 Convergence Check

The iterative updates continue until beliefs stabilize. Stability is measured using KL divergence between the previous and updated belief distributions:

$$D_{KL}(B_i^{(t+1)} || B_i^{(t)}) < 0.0001 \quad (3.6)$$

When the KL divergence is below the threshold for all agents, the module stops iterating. In the proposed implementation, belief propagation is expected to converge in approximately five iterations for the project demonstration scale.

3.1.7 Module-Level Architecture

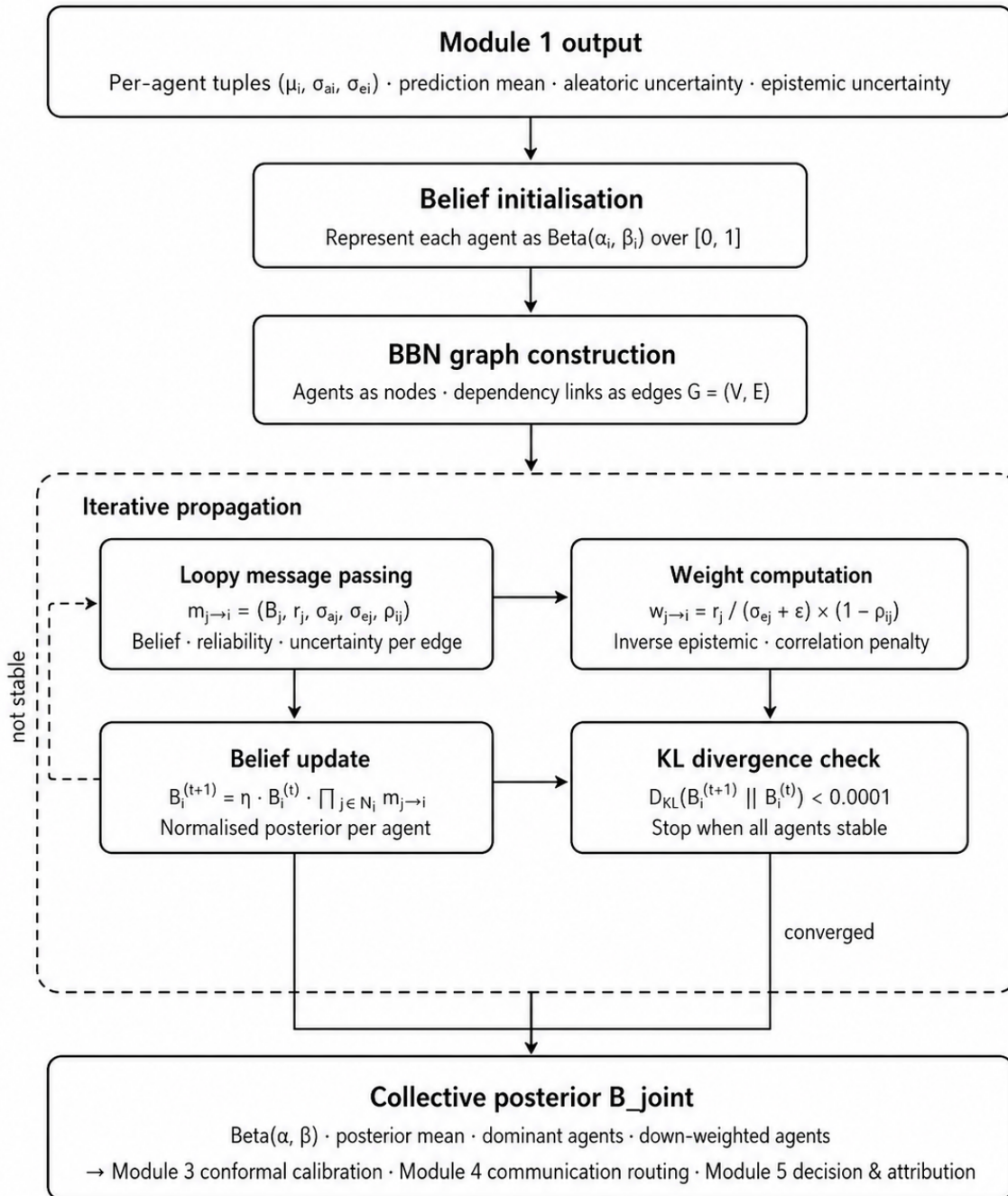


Figure 3.1: Module-level architecture of the Bayesian Belief Network

3.1.8 Overall System Architecture

Within the complete Cognix pipeline, Module 2 sits between uncertainty detection and calibration. It does not replace Module 1; instead, it consumes Module 1's uncertainty decomposition and converts independent agent beliefs into a collective posterior. This posterior then becomes the input for conformal calibration, graph-based communication optimization, and final decision attribution.

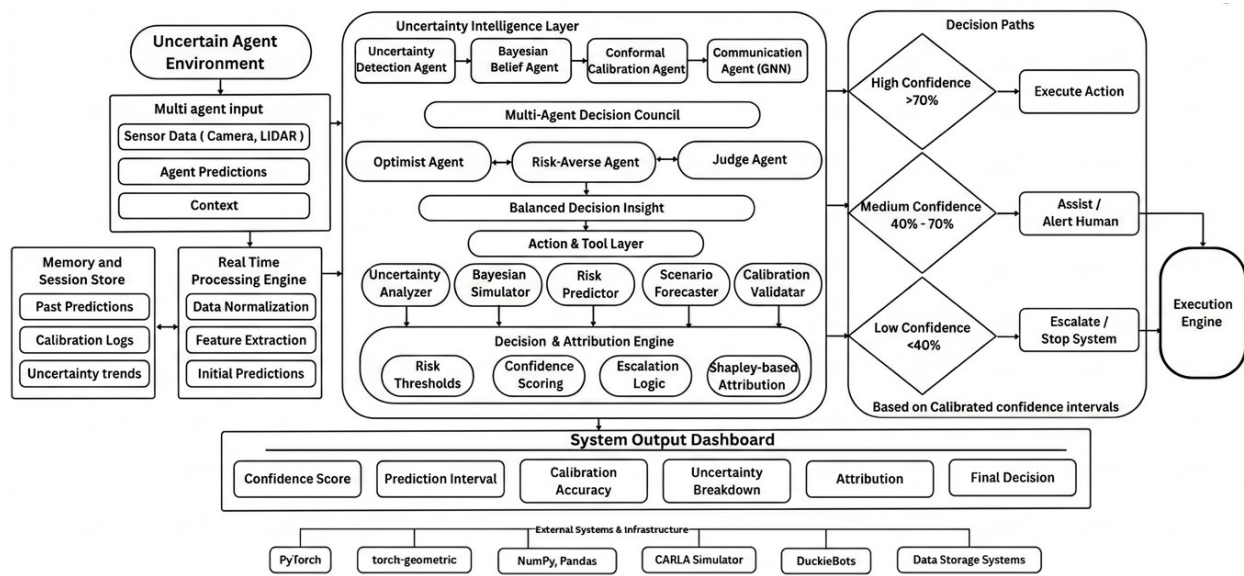


Figure 3.2: Overall Cognix architecture showing the role of Module 2

3.1.9 Algorithm

Algorithm 1: Epistemic-Weighted Loopy Belief Propagation

Input: Agent beliefs B_i , uncertainty tuples $(\sigma_{a,i}, \sigma_{e,i})$, graph $G = (V, E)$

Output: Collective posterior belief B_{joint}

Initialize each node belief using $\text{Beta}(\alpha_i, \beta_i)$;

Construct neighbour set $N(i)$ for each agent;

for each iteration t **do**

for each edge (j, i) **do**

 Compute reliability-adjusted weight $w_{j \rightarrow i}$;

 Send message $m_{j \rightarrow i}$ containing belief and uncertainty context;

for each agent i **do**

 Update belief using weighted incoming messages;

 Normalize updated belief distribution;

 Compute KL divergence between previous and updated beliefs;

if all KL divergence values are below 0.0001 **then**

 stop iteration;

Compute and return collective posterior B_{joint} ;

3.1.10 Tools and Technologies

Table 3.1: Technology stack for Module 2

Tool	Purpose
Python	Core implementation language
NumPy / SciPy	Probability distributions, KL divergence, and numerical computation
pgmpy / PyAgrum	Bayesian Network representation and probabilistic inference support
NetworkX	Agent graph construction and dependency analysis
PyTorch	Integration with Module 1 outputs and tensor-based inference pipeline
Matplotlib / Plotly	Visualizing belief distributions and convergence behaviour

3.1.11 Real-Time Feasibility

The full Cognix system targets a 100ms decision budget. Module 2 is designed to remain lightweight by limiting message passing to the active agent graph, using compact belief distributions, and stopping iterations once KL divergence reaches the convergence threshold. Since the project-scale graph is expected to converge within around five iterations, the module can produce a posterior quickly enough for downstream calibration and decision generation.

Chapter 4

Outcome

4.1 Type of Project

The Cognix project, and specifically Module 2 – the Bayesian Belief Network – is primarily research-oriented with a strong application component. It applies probabilistic graphical modelling to the practical problem of collaborative decision making under uncertainty.

- **Research-Oriented:**

Module 2 addresses a focused research gap: existing Bayesian uncertainty methods generally quantify uncertainty within a single model, and existing Bayesian Belief Network applications often target a specific domain such as smart grids or healthcare. They do not provide a real-time multi-agent belief fusion mechanism that uses epistemic uncertainty as a reliability signal, performs iterative message passing, handles correlated evidence, and outputs a collective posterior for downstream calibration. The module contributes an uncertainty-aware belief propagation layer for multi-agent AI systems.

- **Application-Oriented:**

- ◇ **Autonomous Vehicles:** Multiple vehicles or sensors may disagree about obstacle risk. Module 2 fuses these beliefs by trusting low-epistemic-uncertainty agents more strongly.
- ◇ **Medical Diagnosis:** Different diagnostic agents such as imaging, pathology, and clinical-history models may produce different disease probabilities. Module 2 combines them into one explainable posterior risk estimate.
- ◇ **Smart Grid and Industrial Systems:** Distributed sensors may detect abnormal behaviour under uncertain conditions. Module 2 can model parameter dependencies and identify which beliefs should influence the system posterior.

- **System Integration:**

Module 2 is a central integration point in Cognix. It consumes Module 1 uncertainty tuples and produces the posterior required by Module 3, Module 4, and Module 5. Therefore, its output quality directly affects calibration, communication routing, and final decision reliability.

4.2 Expected Outcomes

4.2.1 Input and Output Format

The input to Module 2 is a set of agent-level belief and uncertainty outputs from Module 1:

Input:

```
[
  {
    agent_id: 1,
    prediction_mu: 0.70,
    sigma_aleatoric: 0.12,
    sigma_epistemic: 0.05
  },
  {
    agent_id: 2,
    prediction_mu: 0.40,
    sigma_aleatoric: 0.10,
    sigma_epistemic: 0.28
  },
  {
    agent_id: 3,
    prediction_mu: 0.85,
    sigma_aleatoric: 0.08,
    sigma_epistemic: 0.04
  }
]
```

The expected output is a collective posterior distribution and supporting convergence information:

Output:

```
{
  collective_posterior_mean: 0.78,
  posterior_distribution: "Beta(alpha=18.6, beta=5.2)",
  convergence_iterations: 5,
  final_kl_divergence: 0.00008,
  dominant_reliable_agents: [1, 3],
}
```

```

    downweighted_agents: [2]
}

```

This output is passed to Module 3 for conformal calibration and to Module 5 for risk-aware action selection and attribution.

4.2.2 Quantitative Targets

Table 4.1: Expected quantitative outcomes for Module 2

Metric	Target
Belief Fusion Accuracy	$> 85\%$
Convergence Iterations	≤ 5 average iterations
Final KL Divergence	< 0.0001
Conflict Resolution Improvement over Averaging	$> 15\%$
Correlation Double-Counting Reduction	measurable reduction in redundant influence
Module Latency	within assigned real-time system budget

4.2.3 Qualitative Contributions

- A working Bayesian Belief Network module that accepts agent uncertainty tuples and returns a collective posterior.
- A belief fusion method that gives higher influence to reliable, low-epistemic-uncertainty agents.
- A correlation-aware mechanism to reduce redundant evidence and avoid overconfident posterior formation.
- A convergence-controlled inference process using KL divergence.
- A clean handoff from uncertainty detection to conformal calibration and final decision making.

4.2.4 Expected Demonstration Result

In the motivating example, simple averaging of agent beliefs 70%, 40%, and 85% gives a final belief of 65%. Module 2 is expected to produce a more meaningful posterior because Agent B’s high epistemic uncertainty reduces its influence, while Agents A and C contribute more strongly. The resulting posterior should move closer to the reliable agents’ belief range while still preserving uncertainty information for calibration and decision making.

Chapter 5

Project Plan

5.1 Project Timeline

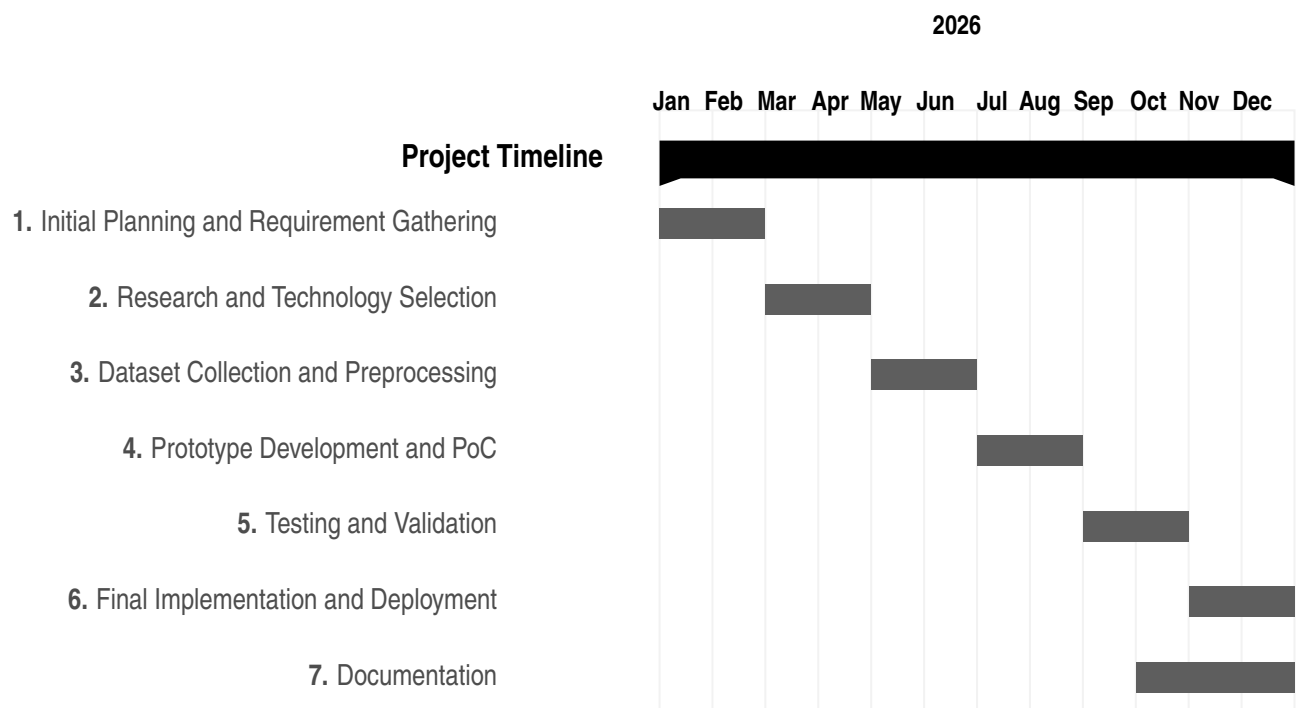


Table 5.1: Project Timeline

5.2 Team Organisation

The Cognix project is developed by a team of four members, each responsible for specific modules of the complete system.

5.2.1 Team Structure

1. Komal — Module 1: Uncertainty Detection Engine

Komal is responsible for designing and implementing the Bayesian Neural Network-based uncertainty decomposition pipeline that forms the foundation of the entire Cognix system.

Key Responsibilities:

- ◊ Aleatoric and Epistemic Uncertainty Decomposition using BNN
- ◊ Variational Inference training with ELBO loss
- ◊ Posterior sampling pipeline (S=50 weight samples)
- ◊ Structured output tuple generation (μ, σ_a, σ_e)
- ◊ CARLA simulation scenario design for validation
- ◊ DuckieBot hardware latency testing and optimisation
- ◊ Literature review for uncertainty detection methods

2. Aryan — Module 2: Bayesian Belief Network

Aryan is responsible for multi-agent probabilistic belief fusion using Loopy Belief Propagation, weighted by epistemic uncertainty from Module 1.

Key Responsibilities:

- ◊ Bayesian Belief Network design and implementation
- ◊ Loopy Belief Propagation algorithm
- ◊ Epistemic-uncertainty-weighted message passing
- ◊ Convergence testing and collective posterior computation

3. Aditya — Module 3: Conformal Calibration Engine

Aditya is responsible for providing statistical guarantees on prediction reliability through conformal calibration techniques.

Key Responsibilities:

- ◊ Conformal Prediction implementation with coverage guarantees
- ◊ Calibration of uncertainty outputs from Module 1 and Module 2
- ◊ Two-stage hierarchical calibration (individual and collective)
- ◊ Non-conformity score computation and threshold selection

4. Zaid — Module 4 and 5: Communication and Decision Engine

Zaid is responsible for efficient inter-agent uncertainty communication and final risk-aware decision making with attribution.

Key Responsibilities:

- ◊ Graph Attention Network for uncertainty-aware communication
- ◊ Bandwidth-constrained message selection protocol
- ◊ Task-specific decision threshold design
- ◊ Shapley value-based uncertainty attribution per agent
- ◊ Escalation decision tree implementation
- ◊ Final decision output formatting and logging

Chapter 6

Experimental setup

In order to evaluate the performance of the COGNIX system, a controlled experimental setup was designed. The aim was to analyze not only prediction accuracy but also the reliability and calibration of outputs. The system was tested under different conditions to understand how well it handles uncertainty in a multi-agent environment. The following sections describe the dataset, technologies used, and performance parameters considered during evaluation.

6.1 Data Set

Synthetic and Benchmark Dataset

- **Source:** Combination of synthetically generated data and standard machine learning benchmark datasets.
- **Formats:** Structured numerical data along with simulated observations representing different agent inputs.
- **Content Variety:**
 - ◊ Regression-based data for continuous prediction tasks
 - ◊ Classification scenarios with probabilistic outputs
 - ◊ Simulated multi-agent observations with varying uncertainty levels

6.2 Technology Used

- **Machine Learning Framework:**
 - ◊ PyTorch for building and training predictive models
- **Uncertainty Estimation:**
 - ◊ Monte Carlo Dropout and ensemble-based techniques for estimating uncertainty
- **Calibration Layer:**

-
- ◊ Conformal Prediction implemented as a post-processing step for generating calibrated intervals
 - **Backend & Processing:**
 - ◊ Python 3.x for overall system implementation
 - ◊ NumPy and Pandas for data manipulation and processing
 - **Development Tools:**
 - ◊ Git for version control and collaboration
 - ◊ Jupyter Notebook for experimentation and analysis

6.3 Performance Parameters

To evaluate the effectiveness of the system, both prediction performance and calibration quality were measured. The following parameters were considered:

- **Prediction Accuracy:**
 - Measures how close the predicted values are to actual values
 - Used standard metrics such as Mean Squared Error (MSE) or accuracy for classification
- **Calibration Accuracy:**
 - Evaluates how well the predicted confidence matches actual correctness
 - Goal: Confidence levels should align with real-world outcomes
- **Coverage:**
 - Percentage of true values that lie within the predicted confidence interval
 - Target: Coverage close to predefined confidence level (e.g., 95%)
- **Prediction Interval Width:**
 - Measures the size of the confidence interval
 - Narrower intervals with correct coverage indicate better performance
- **Computation Time:**
 - Time required to generate predictions and calibration intervals
 - Evaluated to ensure system efficiency
- **System Reliability:**
 - Measures consistency of predictions across different agents
 - Helps in evaluating robustness of multi-agent decision-making

Chapter 7

PO & PSO Mapping

7.1 Program Outcome (PO) Mapping

PO	Description	Level	Justification
PO1	Engineering Knowledge: Apply knowledge of mathematics, science, and engineering fundamentals to AI and DS problems.	3	Cognix applies advanced mathematics including Bayesian inference, probability distributions, Variational Inference, ELBO loss, Shapley values, graph theory, and conformal prediction to solve complex multi-agent uncertainty quantification problems.
PO2	Problem Analysis: Identify, formulate, and analyze complex problems using principles of math, natural science, and engineering.	3	The core problem of over-confident AI systems was identified, formally defined, and analyzed through a review of three recent IEEE papers, reaching the substantiated conclusion that aleatoric and epistemic uncertainty must be separated in real-time multi-agent settings.

PO3	Design/Development of Solutions: Design system components or processes considering societal, safety, and environmental factors.	3	A complete five module system was designed addressing public safety, autonomous driving, medical diagnosis, financial trading with appropriate consideration for regulatory compliance including ISO 26262 and EU AI Act 2.0.
PO4	Conduct Investigations of Complex Problems: Conduct research based investigations, design experiments, analyze data, and synthesize conclusions.	3	Experiments are designed using CARLA simulator across four uncertainty scenario categories, with analysis of decomposition accuracy, ECE, latency, and OOD detection all grounded in research based investigation methodology.
PO5	Modern Tool Usage: Create and apply modern tools, modeling, and engineering techniques.	3	PyTorch 2.0.1, Pyro, BayesianTorch, CARLA 0.9.14, PettingZoo, DuckieBot, PyTorch Geometric, TorchScript, and Weights and Biases are all applied extensively across the Cognix system.
PO6	The Engineer and Society: Apply contextual knowledge to societal, health, safety, legal, and cultural issues.	3	Cognix directly addresses societal safety, autonomous vehicle accidents, medical AI misdiagnosis, financial flash crashes and aligns with legal regulatory requirements of ISO 26262, EU AI Act 2.0, and FDA 2026 guidelines.

PO7	Ethics: Apply ethical principles and commit to professional ethics and responsibilities.	3	Cognix is fundamentally motivated by the ethical responsibility of deploying AI in safety-critical domains. The system ensures AI is transparent and honest about its confidence, not overconfident before making decisions affecting human lives.
PO8	Individual and Team Work: Function effectively as an individual and in teams.	3	Cognix is developed by a four member team with each member owning one module functioning effectively both individually and collaboratively in a multi-disciplinary AI and robotics project.
PO9	Communication Skills: Communicate effectively on complex engineering activities.	3	Technical documentation, IEEE literature review, seminar report, and PPT presentation were all prepared, communicating complex uncertainty quantification concepts clearly to both technical and non-technical audiences.
PO10	Project Management and Finance: Apply project management and financial principles in engineering work.	3	The project was structured across eight phases with clear module-wise division of work, timelines, integration milestones, and hardware procurement planning for DuckieBot and CARLA infrastructure.

PO11	Life-long Learning: Recognize the need for life-long learning amidst technological change.	3	The team independently studied cutting edge research from 2022 to 2025 on Bayesian Neural Networks, conformal prediction, and graph-based uncertainty communication all topics beyond standard curriculum demonstrating self directed life long learning.
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Table 7.1: Program Outcomes (PO) Mapping – Cognix

7.2 Program Specific Outcome (PSO) Mapping

PSO Number	Description	Mapping Level	Justification
PSO1	Interpret data and apply Artificial Intelligence algorithms to provide solutions.	3	Cognix interprets multi-modal sensor data from camera, LIDAR, and radar agents and applies Bayesian Neural Networks, Graph Neural Networks, Loopy Belief Propagation, and conformal prediction algorithms to provide safe and calibrated solutions for multi-agent decision making in autonomous driving, medical diagnosis, and distributed robotics.

PSO2	Apply standard practices, strategies and use appropriate models of data analytics to discover knowledge.	3	Standard Bayesian inference practices, Shapley value attribution analysis, and Expected Calibration Error evaluation strategies are applied to multi-agent sensor data to discover actionable knowledge — identifying which agents are reliable, which uncertainty sources can be reduced, and which scenarios require human intervention.
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Table 7.2: Program Specific Outcomes (PSO) Mapping – Cognix

Chapter 8

Summary

Cognix is designed to solve a problem that existing AI systems largely ignore the inability to explain why they are uncertain and what should be done about it. The system operates across five interconnected modules, where each agent first decomposes its sensor observations into two distinct uncertainty components, fuses its beliefs with neighbouring agents through probabilistic message passing, receives a mathematically guaranteed calibration ensuring stated confidence levels are accurate, communicates only the most informative uncertainty signals to other agents through a learned graph-based protocol, and finally receives a complete decision output that not only recommends an action but also identifies which specific agent is the weakest link in the collective reasoning chain. Applied to autonomous driving, medical diagnosis, and multi-robot coordination, Cognix moves AI decision making away from blind confidence toward a model of structured and explainable self-awareness, where every prediction carries a justified level of trust rather than an arbitrary score.

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